

Predictive Entropy of Multilingual Propaganda: An LLM-Based Analysis

Internship Report

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August 24, 2026

Introduction

Natural language processing (NLP) provides tools for quantifying the complexity and predictability of texts. This report describes a modular NLP pipeline built to process structurally diverse multilingual corpora, including tweets, news articles, and narrative clusters, and applies it to measure the predictive entropy of propaganda in Bits Per Character (BPC) via cross-entropy.

The analysis covers propaganda narratives, sources, and labeling schemas independently within each dataset, and asks how the intent to disseminate propaganda affects language predictability across English and Ukrainian texts. The methodology below builds on classical cognitive experiments and adapts them to establish a ground-truth baseline for evaluating modern language models.

Methodology for Estimating the Entropy Rate of Natural Languages

1. Theoretical Foundations

In information theory, the entropy of a natural language quantifies the underlying complexity and unpredictability of its character sequences [1, 2]. For a stochastic process X , the Shannon entropy $H(X_1^n)$ of a sequence of length n is defined as [1]:

$$H(X_1^n) = - \sum P(X_1^n) \log P(X_1^n)$$

The entropy rate h , which represents the average amount of information per character in an infinitely long sequence, is expressed as [1]:

$$h = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1^n)$$

Because calculating the exact probability distribution of a natural language is computationally infeasible, cognitive experiments based on Claude Shannon’s 1951 methodology [8] are utilized to approximate these bounds [1, 2].

The predictive complexity of the n -th character given the preceding $n - 1$ characters is denoted as $F_n \equiv H(X_n|X_1^{n-1})$ [1]. If the stochastic process is stationary, F_n converges to the entropy rate h as n approaches infinity [1]. In a cognitive experimental setting, human subjects predict the next character, and the number of guesses required to find the correct answer is recorded [1, 2]. Let q_i^n denote the probability that a subject requires exactly i guesses to identify the correct character following a context of length $n - 1$ [1].

Shannon established the following inequality to define the bounds of F_n [1]:

$$\sum_{i=1}^K i(q_i^n - q_{i+1}^n) \log i \leq F_n \leq - \sum_{i=1}^K q_i^n \log q_i^n$$

Here, K represents the total number of characters in the language’s alphabet including the space symbol [1]. For the English language, $K = 27$ [1]. For the Ukrainian language, $K = 34$ to account for the 33 letters of the alphabet plus a whitespace character [2]. Alternatively, the lower bound can be defined as the entropy of a ”least surprising” distribution, calculated as [2]:

$$H_{lower} = \sum_{i=1}^K q_i \log_2 i$$

2. Experimental Design (Cognitive Approach)

To evaluate the entropy rate empirically, large-scale cognitive experiments were designed for both English and Ukrainian, utilizing different data-gathering strategies [1, 2].

- **English Language:** Data was collected via Amazon Mechanical Turk (AMT), gathering 172,954 character predictions from 683 subjects [1]. Participants were asked to guess the subsequent character sequentially from the beginning of the phrase [1].
- **Ukrainian Language:** The experiment relied on 184 volunteers recruited through social media, yielding 17,023 correct observations [2]. To minimize participant frustration and early quitting, subjects were provided with an initial context of 70 characters and evaluated on sentences ranging from 120 to 200 characters in length [2].

3. Data Processing and Mathematical Extrapolation

Due to the high variance in per-position entropy values and the presence of underperforming participants in uncontrolled online settings, both methodologies employ rigorous data trimming [1, 2].

- In the English study, finding the final estimated value required filtering the results to the top 1000 best-performing experimental sessions [1]. The upper and lower bounds were then extrapolated to infinite context lengths using Hilberg’s ansatz function [1]:

$$f_1(n) = An^{\beta-1} + h$$

where $\beta < 1$ [1].

- In the Ukrainian study, 65% of the worst-performing sessions were trimmed, and additional binomial outlier detection was used to discard improbably good results to account for possible cheating [2]. Because context beyond $N = 70$ yields only marginal improvements in guess accuracy, the Ukrainian methodology bypassed ansatz extrapolation [2]. Instead, observations were pooled for positions 70 – 110, and a weighted upper bound was calculated using the formula [2]:

$$H_{upper} = \sum_{n=70}^{110} w_n \cdot H_{upper}(n)$$

where the weight w_n represents the proportion of observations at position n [2]:

$$w_n = \frac{N_n}{\sum N}$$

4. Dataset and Annotation Schemas

The analysis draws on several complementary datasets spanning multiple platforms and domains:

1. **HQP Benchmark Dataset (Twitter):** Comprising 30,000 English tweets collected at the onset of the full-scale Russian war in Ukraine [4], this corpus captures the informal, abbreviated nature of short-form social media discourse. For comparative taxonomy analysis, a validated subset of 500 tweets was evaluated across two distinct schemas: the established span-level technique taxonomy by Sahitaj et al. (2025) [3] (clustered into six high-level functional categories) and a novel intent-focused taxonomy (17 fine-grained labels grouped into 5 strategic intents, such as "*Shift Blame and Justify Aggression*" and "*Confuse and Distract*") [4].
2. **Telegram Channels Corpus:** A large-scale collection of Ukrainian and Russian Telegram posts (1.35 million raw messages, filtered to an evaluation set of over 81,000 messages) gathered from 98 curated war-relevant channels [5]. This domain captures unmoderated wartime communications that combine native forwarding chains with semantic similarity graphs to track narrative diffusion [5].
3. **Scraped News Corpora:** Multilingual news articles systematically scraped from prominent Ukrainian, Russian, Romanian, French, and English media outlets. This dataset reflects formal, institutional journalistic frameworks and is divided into pro-Western/pro-Ukrainian sources and official pro-Kremlin government positions [6].
4. **Labeled News Datasets:** Auxiliary corpora annotated with fine-grained binary propaganda labels and specific keyword dictionaries to evaluate lexical and structural correlations with predictive entropy [6].

5. Data Preprocessing and Evaluation Pipeline

The raw data was structurally diverse, so a preprocessing pipeline was built to standardize it before running the language models.

5.1. Text Cleaning and Normalization

The initial corpus consisted of a propaganda dataset merged with a sampled baseline. The text normalization process handled nested structures (e.g., JSON-like strings and conversational roles) by extracting the raw textual content [7]. The text was then standardized by removing URLs, user mentions, zero-width characters, and HTML tags [7]. Hashtags were dynamically split from CamelCase formats into separate, readable words [7]. Furthermore, domain-specific artifacts, such as Telegram message headers, timestamps, and date-line leads (e.g., "Reuters"), were stripped using regular expressions [7]. To prevent unusually long texts from skewing the token-level analysis, documents were strictly truncated to a maximum length of 4,000 characters [7].

5.2. Boilerplate Removal and Deduplication

Because repetitive non-semantic text artificially lowers entropy, extensive boilerplate removal was implemented across three distinct layers [7]:

- **Static Pattern Matching:** A comprehensive list of regular expressions removed known call-to-action (CTA) phrases, subscription prompts, and outlet slogans [7].
- **Structural Cleansing:** Embedded ALL-CAPS headlines, frequently spliced mid-paragraph in certain news formats, were detected and removed structurally (defined as 5 or more consecutive capitalized Latin words) [7].
- **Statistical Deduplication:** Both intra-document and cross-document repetition filters were applied [7]. Sentences repeating 3 or more times within the same document were excised [7]. Additionally, a generic boilerplate stripper removed sentences that recurred across multiple documents within the same source file, prioritizing strings positioned near the edges (headers/footers) [7].

Following the cleaning phase, quality filters discarded texts with an alphabetic-character ratio below 0.5 and lengths under 100 characters [7]. Language detection was applied to retain only Ukrainian (*uk*) and English (*en*) texts [7]. Finally, global deduplication was enforced using exact MD5 hashing alongside near-duplicate detection via MinHash and Locality-Sensitive Hashing (LSH) with 128 permutations and a Jaccard similarity threshold of 0.90 [7].

5.3. Sentence Explosion and Context Formatting

To provide a consistent evaluation window for the language models, the deduplicated documents were "exploded" into discrete sentences separated by terminal punctuation [7]. Only

sentences with lengths strictly between 120 and 200 characters were retained for the final evaluation dataset [7].

6. LLM Deployment and Entropy Calculation

6.1. Model Portfolio and Quantization Strategy

The predictability of the processed text was measured using a diverse suite of causal language models [7]:

- google/gemma-3-1b-pt
- meta-llama/Llama-3.2-3B
- Qwen/Qwen2.5-7B-Instruct
- google/gemma-3-12b-pt
- lapa-llm/lapa-12b-pt

To accommodate hardware memory constraints, models with 7 billion parameters or more (Qwen2.5, Gemma-3-12B, and Lapa-12B) were loaded using 4-bit quantization via the `BitsAndBytes` library [7]. Specifically, the models utilized the `NormalFloat4` (`nf4`) quantization type with a `bfloat16` compute data type [7]. *Impact of Quantization:* 4-bit quantization reduces the VRAM footprint enough to run the larger models on consumer-grade GPUs, at the cost of some precision loss in the model’s logits. For aggregate cross-entropy over large corpora this distortion turned out to be negligible in practice, so quantization was used for all models with 7B+ parameters.

Table 1: Model portfolio used for BPC calculation.

Model	Parameters	Quantization	Compute dtype
google/gemma-3-1b-pt	1B	None (full precision)	bfloat16
meta-llama/Llama-3.2-3B	3B	None (full precision)	bfloat16
Qwen/Qwen2.5-7B-Instruct	7B	4-bit (nf4)	bfloat16
google/gemma-3-12b-pt	12B	4-bit (nf4)	bfloat16
lapa-llm/lapa-12b-pt	12B	4-bit (nf4)	bfloat16

6.2. Calculating Bits Per Character (BPC)

The text sequences were tokenized with offset mappings to maintain precise character-to-token alignment [7]. To address the LLM ”burn-in” phase—where models output artificially high loss on initial tokens due to a lack of prior context—a strict context mask was applied [7]. Using the offset mappings, the cross-entropy loss for any token starting within the first 70 characters of the sentence was ignored (labels set to -100) [7].

During the forward pass under a no-gradient context, the summed cross-entropy loss for the valid (unmasked) tokens was computed in *nats* [7]. This sum was converted to *bits* by dividing by $\ln(2)$, and finally normalized by the number of valid characters (total sequence length minus the 70-character mask) to yield the standardized Bits Per Character (BPC) metric [7]:

$$BPC = \frac{\sum Loss_{nats}}{\ln(2) \cdot N_{valid.chars}}$$

7. Results and Statistical Analysis

The quantitative analysis of predictive entropy (BPC) shows that propaganda does not shift entropy in one consistent direction. Its predictability instead depends on the dissemination platform and the specific rhetorical strategy used.

7.1. The Platform-Dependent Entropy Inversion

The clearest finding is a platform-dependent inversion of propaganda entropy: social media and traditional news move in opposite directions.

On social media (Ukrainian Telegram and English Twitter), propaganda text shows a statistically significant shift toward *higher* entropy than non-propagandistic baselines. The Qwen2.5-7B-Instruct model, for example, measured a mean BPC of 1.389 for Telegram propaganda versus 1.356 for the baseline ($p_{fdr} < 0.001$). This is consistent with propagandistic discourse on social media using more emotive language, varied slang, and unpredictable phrasing aimed at engagement.

On the English Scraped News dataset, this reverses: propaganda has significantly *lower* entropy (0.749 BPC) than standard journalism (0.807 BPC). This is consistent with state-sponsored or institutional propaganda in formal news outlets leaning on a restricted vocabulary and repetitive, coordinated talking points, which makes the text more predictable to a language model.

7.2. Cross-Model Reliability and Statistical Significance

To check that this pattern was not specific to one model, the entropy calculations were replicated across all five architectures (1B to 12B parameters, including quantized variants). All five agreed on the direction of the shift in every group, which points to the BPC differences being a property of the text itself rather than an artifact of a particular tokenizer or model.

Significance was tested with a two-sided Mann-Whitney U test and Benjamini-Hochberg FDR correction. Telegram and Twitter both came back significant at $p_{fdr} < 0.001$, with rank-biserial effect sizes indicating a moderate but consistent class separation. The one exception was Ukrainian Scraped News ($p_{fdr} > 0.05$), which is not surprising given the class imbalance in that set (163 propaganda examples vs. 9,979 baseline).

Table 4 in the Appendix reports the full per-model breakdown underlying this summary.

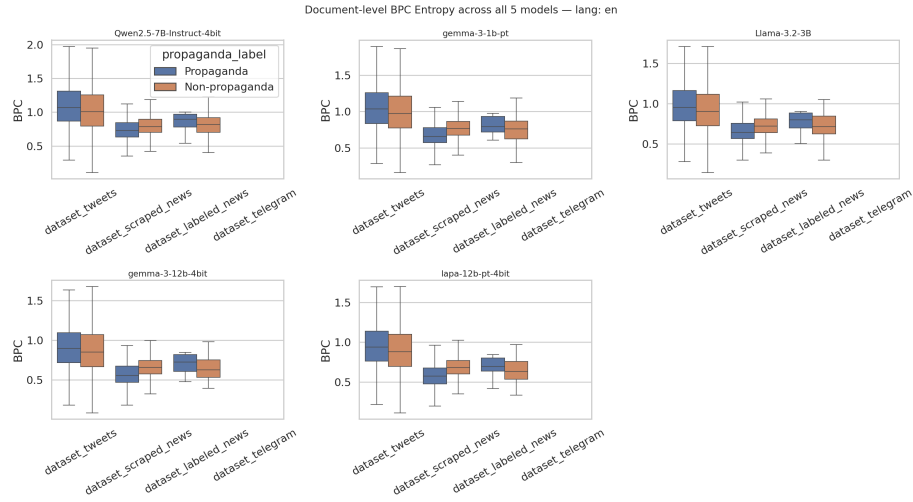


Figure 1: BPC Entropy across 5 models (en)

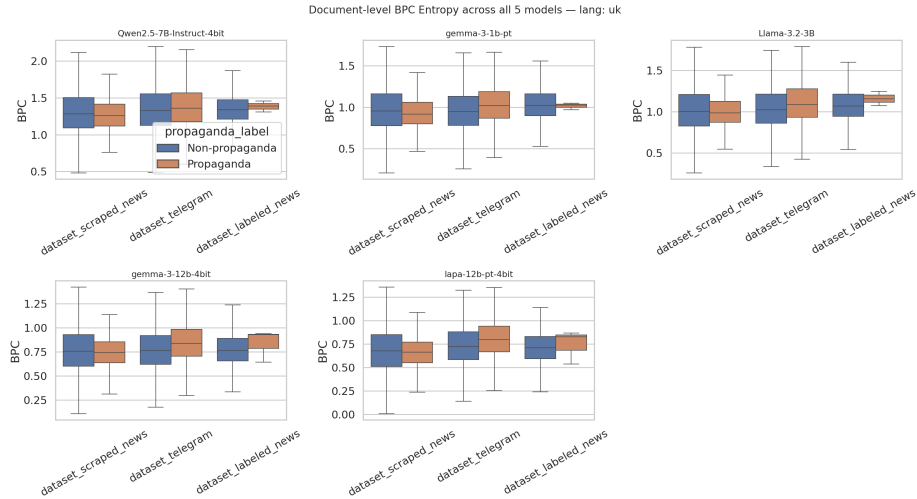


Figure 2: BPC Entropy across 5 models (uk)

Table 2: Cross-model summary of propaganda vs. baseline BPC by dataset. Means are averaged across all five models; p_{fdr} (max) is the least significant of the five per-model FDR-corrected p -values, so “significant” indicates agreement across every model.

Dataset	Lang	Prop. BPC	Base. BPC	Δ	n_{prop}	n_{base}	Sig. (all 5)
English Tweets	en	1.014	0.969	+0.045	1,525	7,879	Yes ($p_{fdr} < 10^{-6}$)
Ukrainian Telegram	uk	1.043	0.980	+0.063	5,368	12,464	Yes ($p_{fdr} < 10^{-10}$)
English Scraped News	en	0.652	0.735	-0.083	107	1,835	Yes ($p_{fdr} < 10^{-3}$)
Ukrainian Scraped News	uk	0.922	0.960	-0.038	163	9,979	No ($p_{fdr} = 0.41$)

7.3. Rhetorical Strategies and Entropy Signatures

Beyond the binary propaganda/non-propaganda split, individual rhetorical strategies produce distinct entropy signatures.

In the English social media data, *slogans* and *name-calling/labeling* register the highest median BPC. This tracks with these strategies drawing on rare derogatory tokens, creative insults, and one-off neologisms. *Bandwagon* appeals and *appeals to fear/prejudice*, by contrast, show noticeably lower median BPC consistent with fear-mongering and consensus-faking relying on recycled, formulaic phrasing (e.g., stock threat scenarios) that a language model finds easy to predict.

In the Ukrainian Telegram data, narratives framing Ukraine as a "terrorist state" produce higher entropy, while economic narratives (e.g., "Supporting Ukraine hurts the West") produce lower entropy – again suggesting a more standardized framing for the latter.

7.4. Linguistic and Lexical Correlates of Entropy

To identify what’s driving predictability at a finer grain, a Spearman correlation was run between document-level BPC and specific linguistic features: keyword frequency, emotional valence, and sentence structure.

Standardized geopolitical terminology and established propaganda clichés (*in ukraine, europe, kremlin, fascism, genocide, fake*) correlate negatively with BPC — texts built around these keywords are more formulaic and predictable. Emotionally charged or context-specific terms (*children, azov, hysteria*) correlate positively, adding statistical surprise.

Structural and emotional features show the same pattern. As shown in Figure 4, average sentence length correlates negatively with BPC: longer sentences in news and propaganda tend toward bureaucratic or pseudolegal boilerplate, which is easy for an LLM to predict.

Fear emotion, on the other hand, correlates positively with BPC. Text written to evoke fear or panic tends to break from rigid templates and use more expressive phrasing, which raises entropy.

7.5. Comparative Analysis with Established Entropy Bounds

To gauge how predictable the LLM outputs are relative to human performance, this section compares the LLM cross-entropy results against the cognitive entropy bounds from the literature.

Ren et al. (2019) established a human cognitive upper bound of $h \approx 1.22$ BPC for English [1]. Table 3 compares this against both the full range of the five-model portfolio and against Qwen2.5-7B-Instruct specifically, since Qwen2.5-7B is consistently the highest-BPC (least predictive) model in the portfolio and the one that comes closest to the human bound. On English Scraped News, even Qwen2.5-7B (0.749 – 0.807 BPC) sits well below the human bound, and the other four models go as low as 0.577 BPC and since formal news text is exactly what these models were trained to predict well. On English Tweets, entropy is higher (0.893 – 1.109 BPC), and Qwen2.5-7B (1.059 – 1.109 BPC) comes closest to the human boundary. This reflects the structural irregularity of social media text.

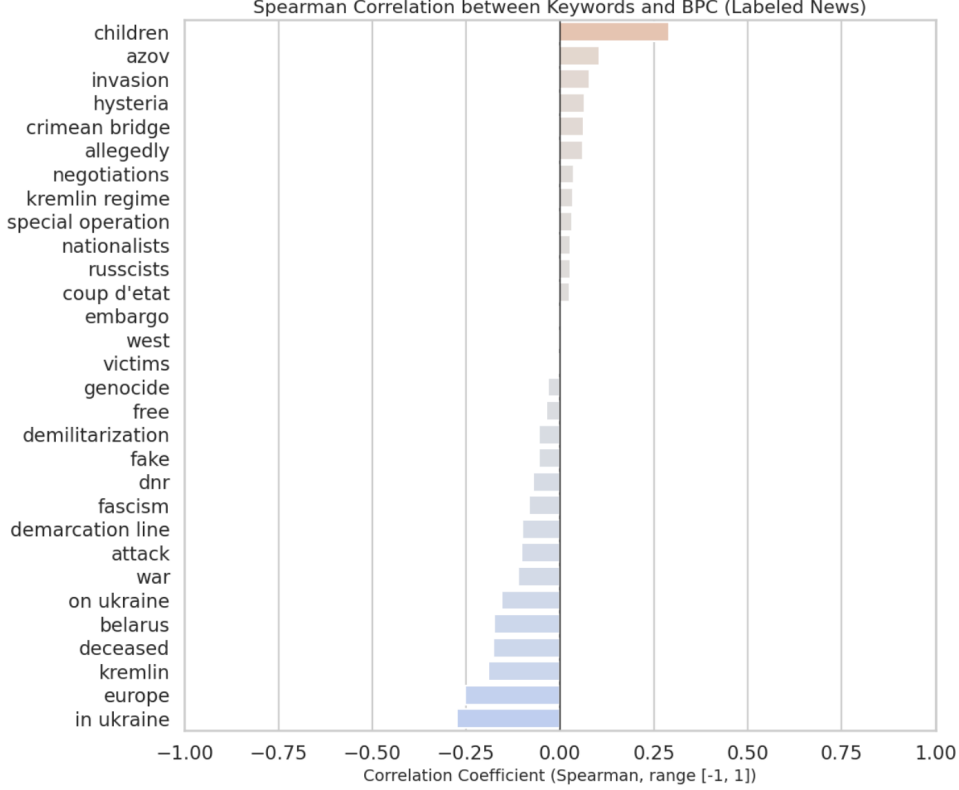


Figure 3: Spearman Correlation between specific keywords and BPC in Labeled News.

The Ukrainian results follow a similar pattern but with a wider spread. Lavreniuk et al. (2026) established an upper cognitive bound of $H_{upper} \approx 1.201$ BPC for standard Ukrainian news, and reported LLM cross-entropy of $0.717 - 0.980$ BPC on similar formal corpora [2]. Ukrainian Scraped News in our data spans $0.664 - 1.317$ BPC across the portfolio; only Qwen2.5-7B ($1.271 - 1.317$ BPC) approaches a genuine challenge, while the other four models stay comfortably below the human bound. Ukrainian Telegram shows the same split: the full-portfolio range ($0.742 - 1.389$ BPC) mostly sits within or below the human bound, but Qwen2.5-7B ($1.356 - 1.389$ BPC) exceeds it [2].

This means the "exceeds the human bound" result is model-specific, not a property of the dataset as a whole. It suggests that Ukrainian Telegram text falls outside Qwen2.5-7B's training distribution specifically – the smaller and more heavily quantized models in the portfolio handle the same text much closer to their usual range. One reading is that wartime Telegram discourse (rapid narrative shifts, dense military jargon, high emotional valence) is a harder target for this particular model than standard journalism is for a human reader.

8. Conclusion

This internship project built a modular NLP pipeline to measure the linguistic predictability of multilingual propaganda, adapting Shannon's cognitive entropy framework to modern LLMs. The main finding is that propaganda's entropy is not fixed in one direction: it

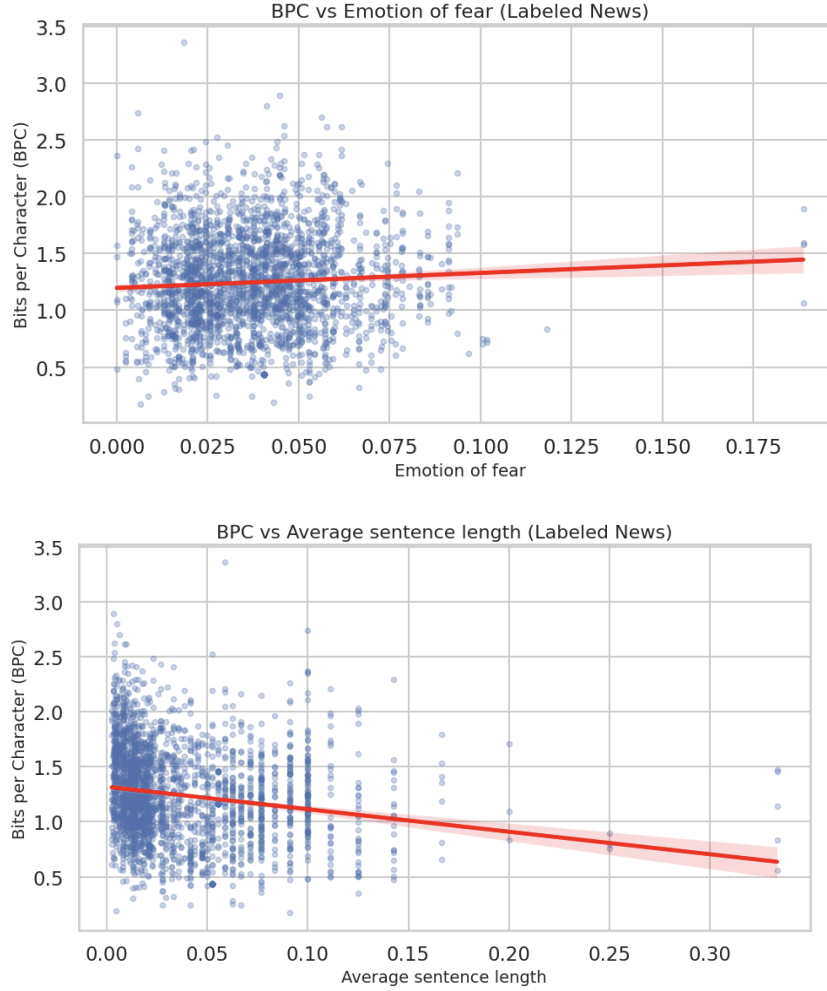


Figure 4: Scatter plots demonstrating the relationship between BPC and Average Sentence Length and Emotion of Fear.

Table 3: LLM cross-entropy (BPC) compared with established human cognitive upper bounds. “Full range” spans all five models in Table 1; Qwen2.5-7B is broken out separately as it is consistently the highest-BPC model in the portfolio.

Language	Domain	Human bound	Full 5-model BPC range	Qwen2.5-7B BPC	
English	Scraped News	$h \approx 1.22$ [1]	0.577 – 0.807	0.749	–
				0.807	
English	Tweets	$h \approx 1.22$ [1]	0.893 – 1.109	1.059	–
				1.109	
Ukrainian	Scraped News	$H_{upper} \approx 1.201$ [2]	0.664 – 1.317	1.271	–
				1.317	
Ukrainian	Telegram	$H_{upper} \approx 1.201$ [2]	0.742 – 1.389	1.356	–
				1.389	

depends on where the text is published. Social media propaganda is more chaotic (higher entropy) than the baseline, while news propaganda is more formulaic (lower entropy) – a platform-dependent inversion that gives a quantitative angle on how digital manipulation differs by medium. The entropy signatures tied to specific rhetorical strategies further suggest that predictive modeling can pick up not just whether a text is propaganda, but something about the intent behind it.

Appendix A: Full Per-Model Statistical Test Results

Table 4: Per-model Mann-Whitney U test results for propaganda vs. baseline BPC, all five models and four dataset/language groups (English Labeled News excluded; see Table 2 for the cross-model summary).

Dataset	Model	Mean BPC (prop.)	Mean BPC (base.)	Rank-biserial	p_{fdr}
Telegram (uk)	gemma-3-12b-4bit	0.855	0.779	−0.196	8.4×10^{-95}
Telegram (uk)	lapa-12b-pt-4bit	0.814	0.742	−0.188	1.5×10^{-87}
Telegram (uk)	gemma-3-1b-pt	1.041	0.970	−0.161	1.1×10^{-64}
Telegram (uk)	Llama-3.2-3B	1.117	1.051	−0.142	1.0×10^{-50}
Telegram (uk)	Qwen2.5-7B-Instruct	1.389	1.356	−0.067	6.2×10^{-12}
Tweets (en)	lapa-12b-pt-4bit	0.968	0.921	−0.106	1.3×10^{-10}
Tweets (en)	gemma-3-1b-pt	1.073	1.022	−0.106	1.3×10^{-10}
Tweets (en)	Qwen2.5-7B-Instruct	1.109	1.059	−0.097	3.6×10^{-9}
Tweets (en)	Llama-3.2-3B	0.989	0.947	−0.095	6.8×10^{-9}
Tweets (en)	gemma-3-12b-4bit	0.931	0.893	−0.087	1.3×10^{-7}
Scraped News (en)	lapa-12b-pt-4bit	0.587	0.690	+0.390	3.8×10^{-11}
Scraped News (en)	gemma-3-12b-4bit	0.577	0.666	+0.343	4.6×10^{-9}
Scraped News (en)	gemma-3-1b-pt	0.685	0.778	+0.336	8.0×10^{-9}
Scraped News (en)	Llama-3.2-3B	0.662	0.734	+0.274	2.5×10^{-6}
Scraped News (en)	Qwen2.5-7B-Instruct	0.749	0.808	+0.209	3.6×10^{-4}
Scraped News (uk)	gemma-3-1b-pt	0.934	0.986	+0.080	0.101
Scraped News (uk)	Qwen2.5-7B-Instruct	1.271	1.317	+0.053	0.284
Scraped News (uk)	Llama-3.2-3B	0.994	1.036	+0.051	0.295
Scraped News (uk)	lapa-12b-pt-4bit	0.664	0.691	+0.047	0.319
Scraped News (uk)	gemma-3-12b-4bit	0.748	0.772	+0.038	0.407

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Note on AI Assistance: AI assistants were used for LaTeX formatting, proofreading, and code optimization. The core methodology, experimental design, and data analysis are the sole work of the author.